

ANOVA example

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```
library(report)
set.seed(20220622)
knitr::opts_chunk$set(echo = TRUE)
```

ANOVA example

This document demonstrates a simple one-way ANOVA example using simulated data.

Enter data

We simulate two groups and then add a third group for the ANOVA.

```
dt_A <- rnorm(n = 10, mean = 1.0, sd = 0.3)
fct_A <- rep(x = 'A', times = length(dt_A))

dt_B <- rnorm(n = 10, mean = 1.5, sd = 0.3)
fct_B <- rep(x = 'B', times = length(dt_B))

dt <- data.frame(
  value = c(dt_A, dt_B),
  condition = c(fct_A, fct_B)
)

summary(dt)
```

```
##      value      condition
## Min.   :0.6043  Length:20
## 1st Qu.:1.0476   Class :character
## Median :1.3087   Mode  :character
## Mean    :1.2435
## 3rd Qu.:1.5615
## Max.    :1.7001
```

Perform a t-test

We begin with a two-sample Student's t-test on the two initial conditions.

```
tt <- t.test(
  formula = value ~ condition,
  data = dt,
  alternative = "two.sided",
  var.equal = TRUE
)

print(tt)

##
## Two Sample t-test
##
## data: value by condition
## t = -3.2324, df = 18, p-value = 0.004621
## alternative hypothesis: true difference in means between group A and group B is not equal to 0
## 95 percent confidence interval:
## -0.6878178 -0.1459245
## sample estimates:
## mean in group A mean in group B
## 1.035062 1.451933

suppressWarnings(report(tt))

## In the report() function, for htest objects, you
## can try providing the data argument manually,
## e.g., report(x, data = data).

## Effect sizes were labelled following Cohen's (1988)
## recommendations.
##
## The Two Sample t-test testing the difference of
## value by condition (mean in group A = 1.04, mean in
## group B = 1.45) suggests that the effect is
## negative, statistically significant, and large
## (difference = -0.42, 95% CI [-0.69, -0.15], t(18) =
## -3.23, p = 0.005; Cohen's d = -1.52, 95% CI [-2.56,
## -0.46])
```

Add another condition for ANOVA

We add a third condition and merge it into the dataset so that ANOVA can be used.

```
dt_C <- rnorm(n = 10, mean = 1.1, sd = 0.3)
fct_C <- rep(x = 'C', times = length(dt_C))

dt_new <- data.frame(value = dt_C, condition = fct_C)

dt_all <- merge.data.frame(x = dt, y = dt_new, all = TRUE)

summary(dt_all)

##      value      condition
## Min.   :0.5428 Length:30
```

```
## 1st Qu.:0.9018   Class :character
## Median :1.1708   Mode  :character
## Mean    :1.1771
## 3rd Qu.:1.5047
## Max.    :1.7001
```

One-way ANOVA using aov()

Because there are now three conditions, we use `aov()` instead of `t.test()`.

```
av <- aov(value ~ condition, data = dt_all)
av_summary <- summary(av)
print(av_summary)

##              Df Sum Sq Mean Sq F value    Pr(>F)
## condition      2  1.133   0.5667    6.727 0.00426 **
## Residuals     27  2.275   0.0842
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

av_Ftest <- anova(object = av, test = 'F')
print(av_Ftest)
```

```
## Analysis of Variance Table
##
## Response: value
##              Df Sum Sq Mean Sq F value    Pr(>F)
## condition      2 1.1334   0.56672    6.7275 0.004259 **
## Residuals     27 2.2745   0.08424
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

suppressWarnings(report(av))
```

```
## The ANOVA (formula: value ~ condition) suggests that:
##
## - The main effect of condition is statistically
## significant and large (F(2, 27) = 6.73, p = 0.004;
## Eta2 = 0.33, 95% CI [0.08, 1.00])
##
## Effect sizes were labelled following Field's (2013)
## recommendations.
```

ANOVA post-hoc tests with correction

We use `emmeans` to perform pairwise comparisons with Sidak correction.

```
library(emmeans)
av_em <- emmeans(object = av, specs = pairwise ~ condition, adjust = 'sidak')
av_ph <- emmeans::test(object = av_em)
print(av_ph$contrasts)

## contrast estimate    SE df t.ratio p.value
## A - B      -0.41687 0.13 27  -3.212  0.0102
## A - C      -0.00924 0.13 27  -0.071  0.9998
```

```
## B - C      0.40763 0.13 27    3.140  0.0121
##
## P value adjustment: sidak method for 3 tests
```

Use a linear model instead of ANOVA

This section shows how the same question can be answered through `lm()`.

```
lmd <- lm(value ~ condition, data = dt_all)
lmd_Ftest <- anova(object = lmd, test = 'F')
print(lmd_Ftest)
```

```
## Analysis of Variance Table
##
## Response: value
##           Df Sum Sq Mean Sq F value    Pr(>F)
## condition  2 1.1334  0.56672    6.7275 0.004259 **
## Residuals 27 2.2745  0.08424
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
suppressWarnings(report(lmd_Ftest))
```

```
## The ANOVA suggests that:
##
## - The main effect of condition is statistically
## significant and large ( $F(2, 27) = 6.73$ ,  $p = 0.004$ ;
##  $\text{Eta}^2 = 0.33$ , 95% CI [0.08, 1.00])
##
## Effect sizes were labelled following Field's (2013)
## recommendations.
```

```
lmd_summary <- summary(lmd)
print(lmd_summary)
```

```
##
## Call:
## lm(formula = value ~ condition, data = dt_all)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.54313 -0.24756  0.06468  0.16433  0.52875
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.03506    0.09178   11.277 1.01e-11 ***
## conditionB     0.41687    0.12980    3.212  0.0034 **
## conditionC     0.00924    0.12980    0.071  0.9438
## ---
## Signif. codes:
## 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2902 on 27 degrees of freedom
## Multiple R-squared:  0.3326, Adjusted R-squared:  0.2832
## F-statistic: 6.727 on 2 and 27 DF,  p-value: 0.004259
```

Post-hoc tests on the linear model

We perform the same pairwise comparisons using the linear model object.

```
lmd_em <- emmeans(object = lmd, specs = pairwise ~ condition, adjust = 'sidak')
lmd_ph <- emmeans::test(object = lmd_em)
print(lmd_ph$contrast)
```

```
## contrast estimate SE df t.ratio p.value
## A - B      -0.41687 0.13 27  -3.212  0.0102
## A - C      -0.00924 0.13 27  -0.071  0.9998
## B - C       0.40763 0.13 27   3.140  0.0121
##
## P value adjustment: sidak method for 3 tests
```